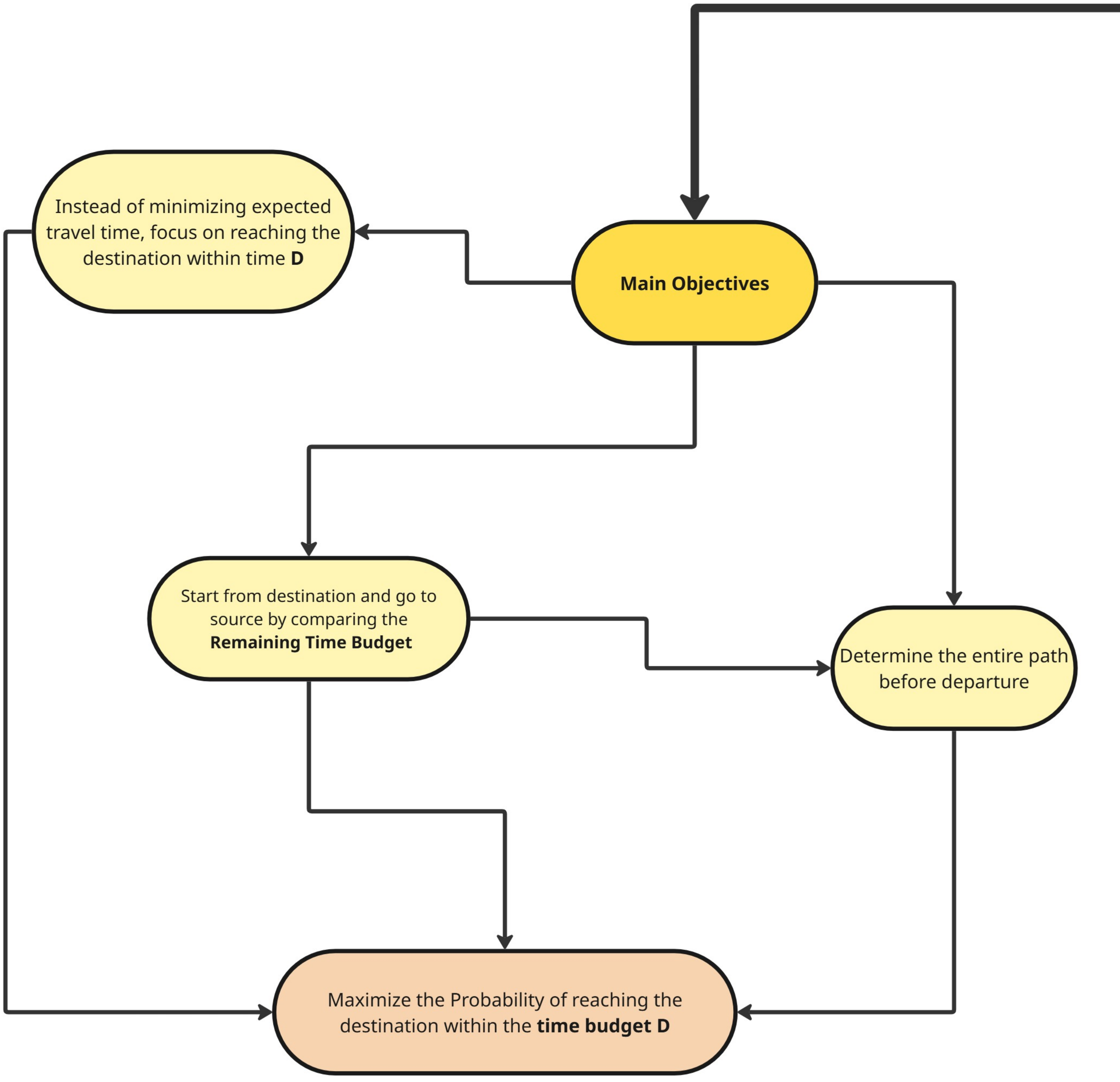
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Determine the entire path before departure
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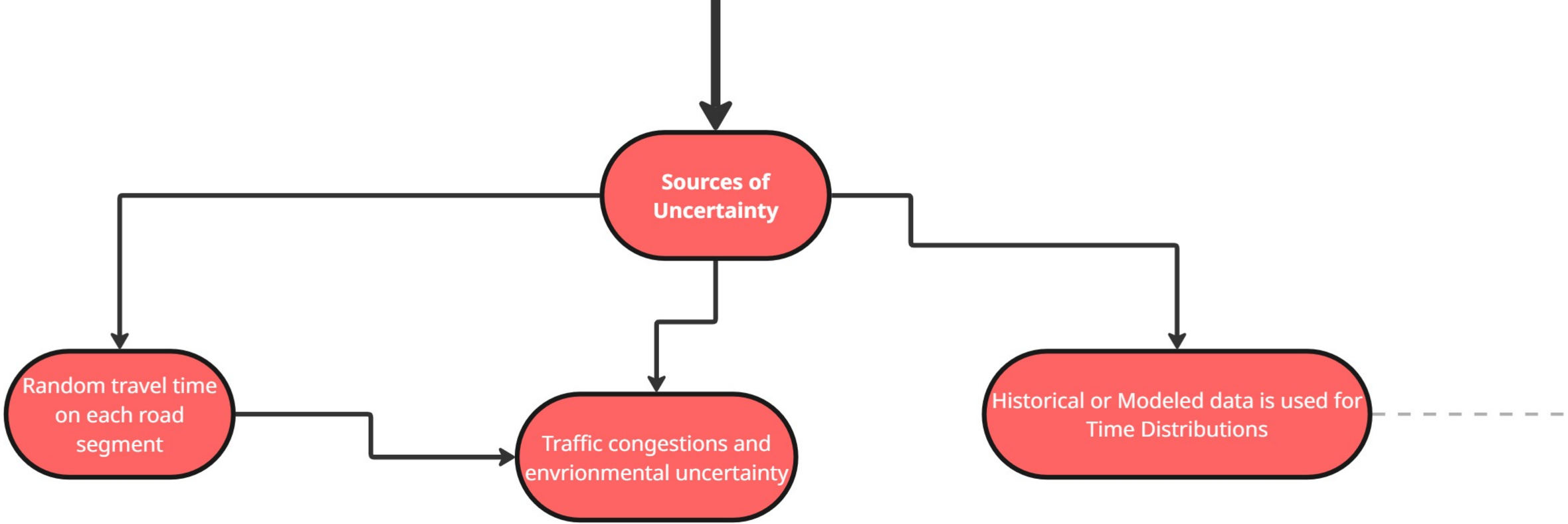
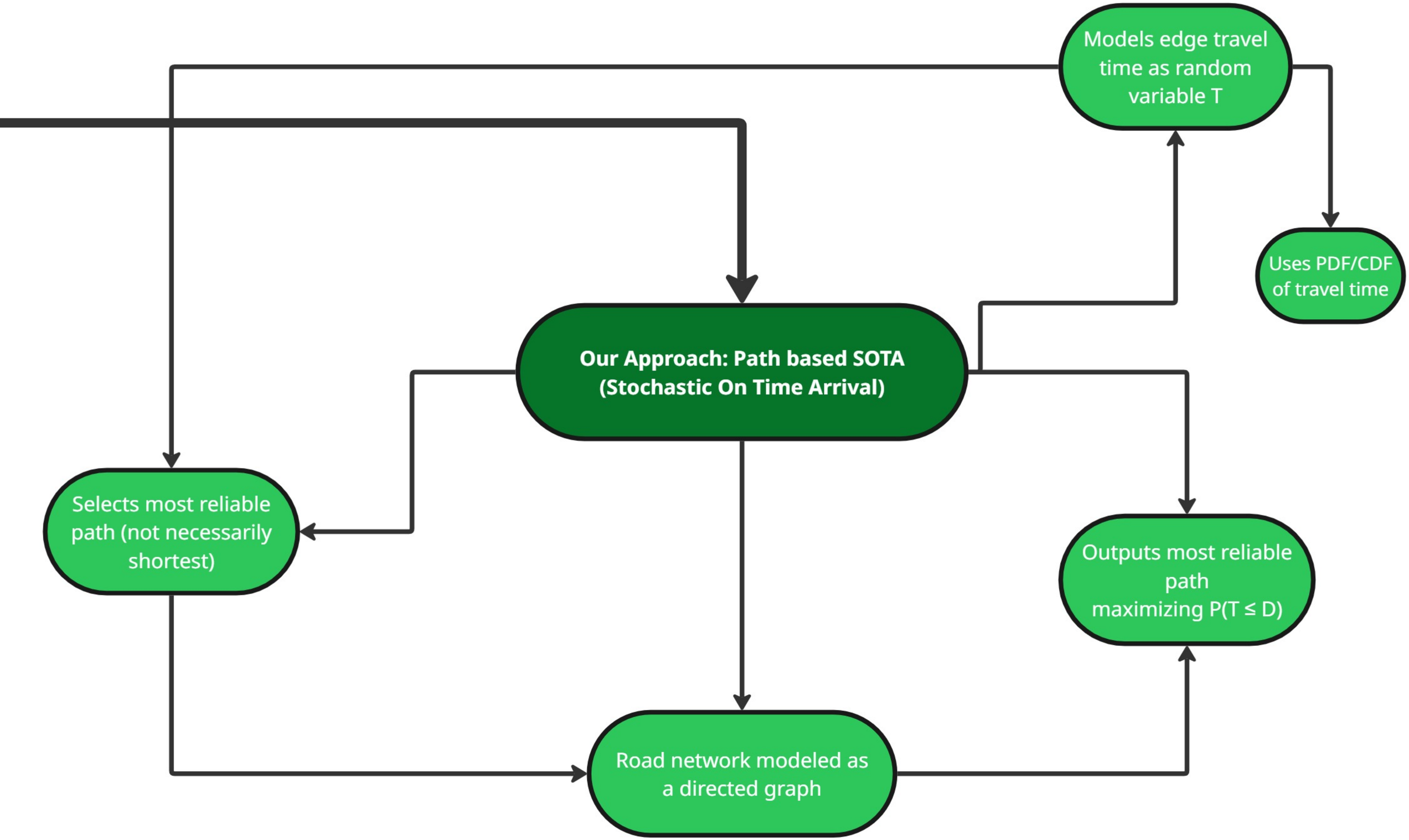

Sources of Uncertainty
Random travel time on each road segment
Traffic congestions and envrionmental uncertainty
Historical or Modeled data is used for Time Distributions


Models edge travel time as random variable T
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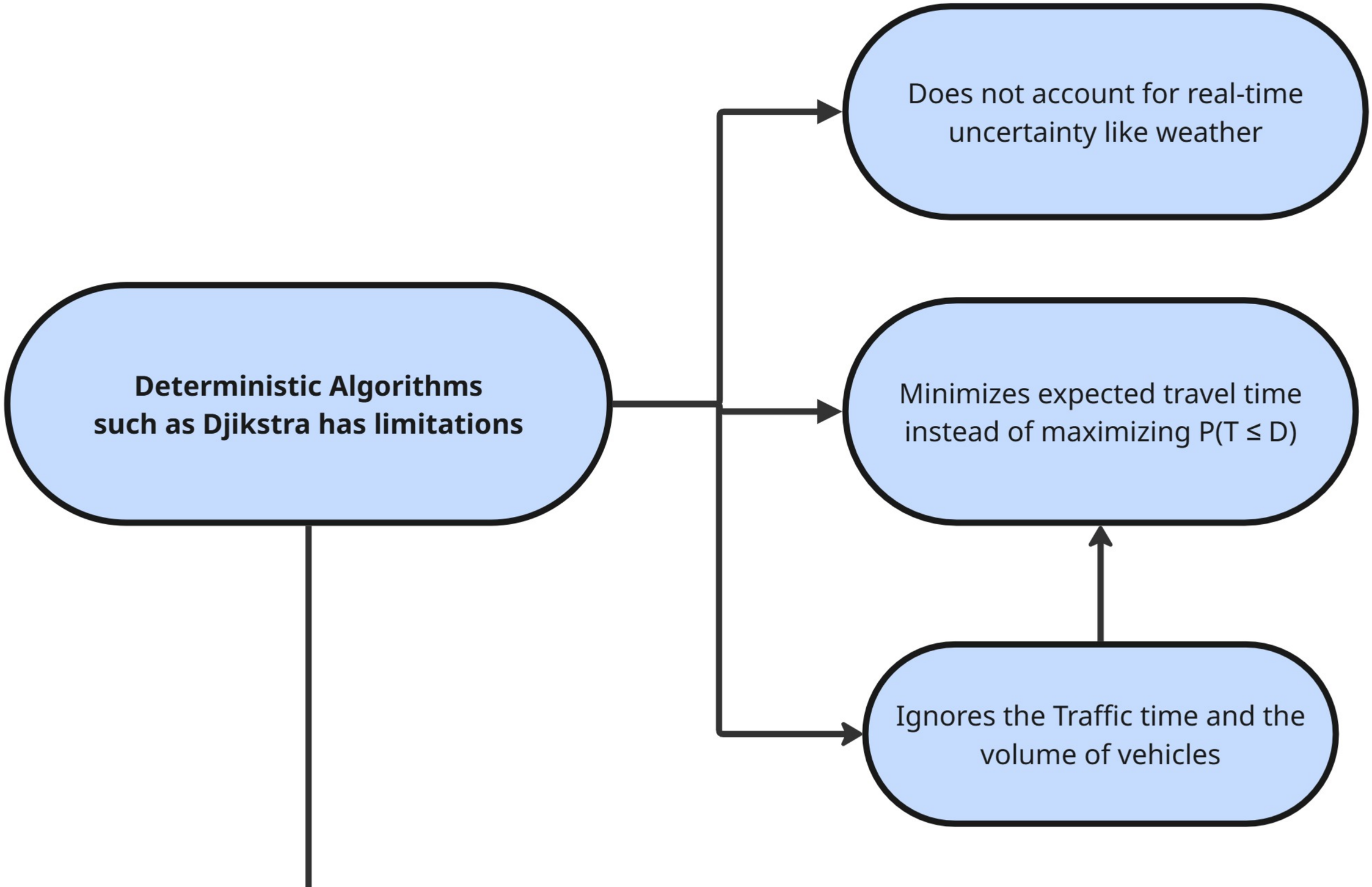
Selects most reliable path (not necessarily shortest)
Outputs most reliable path maximizing P(T ≤ D)
Road network modeled as a directed graph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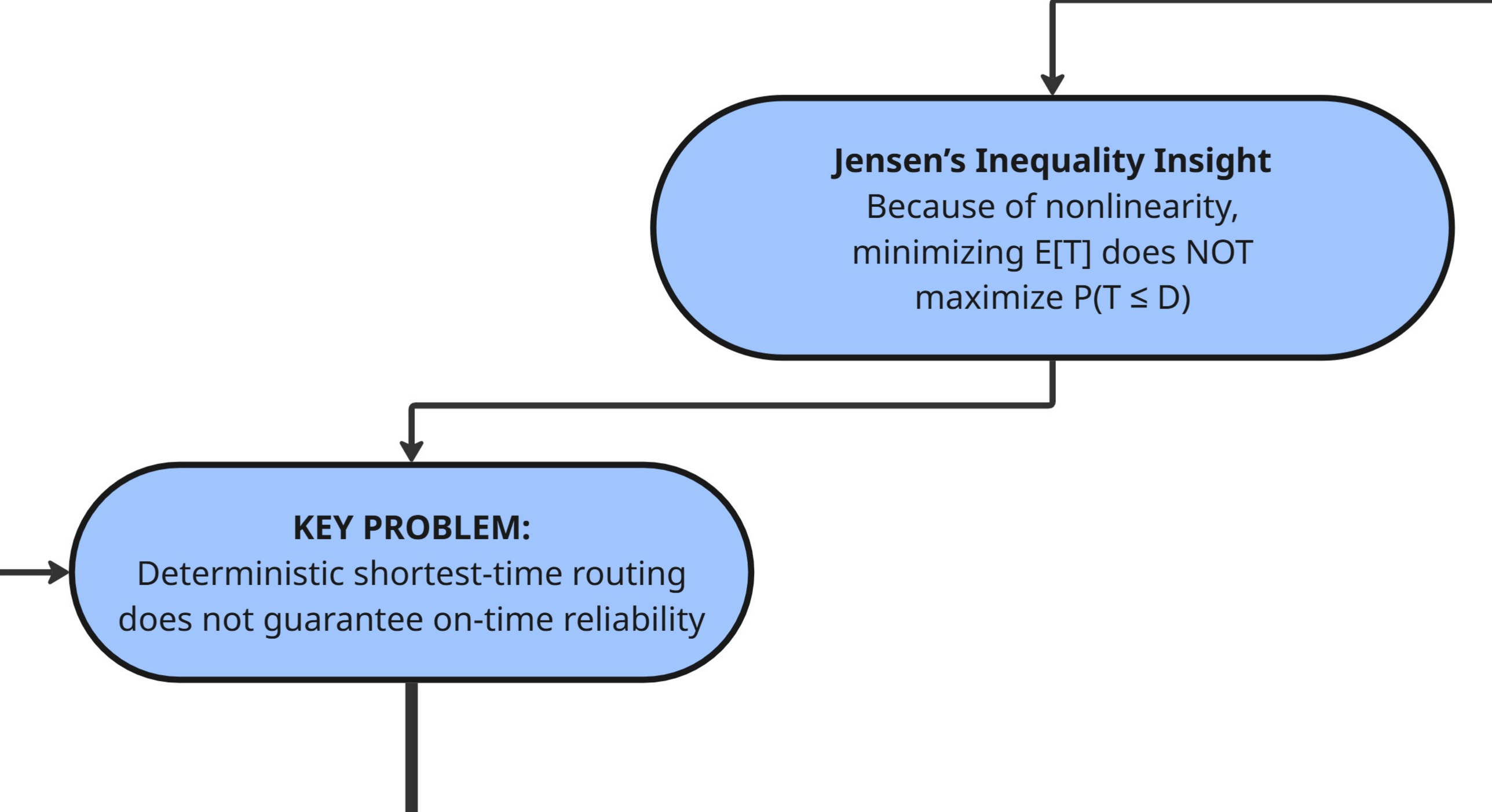
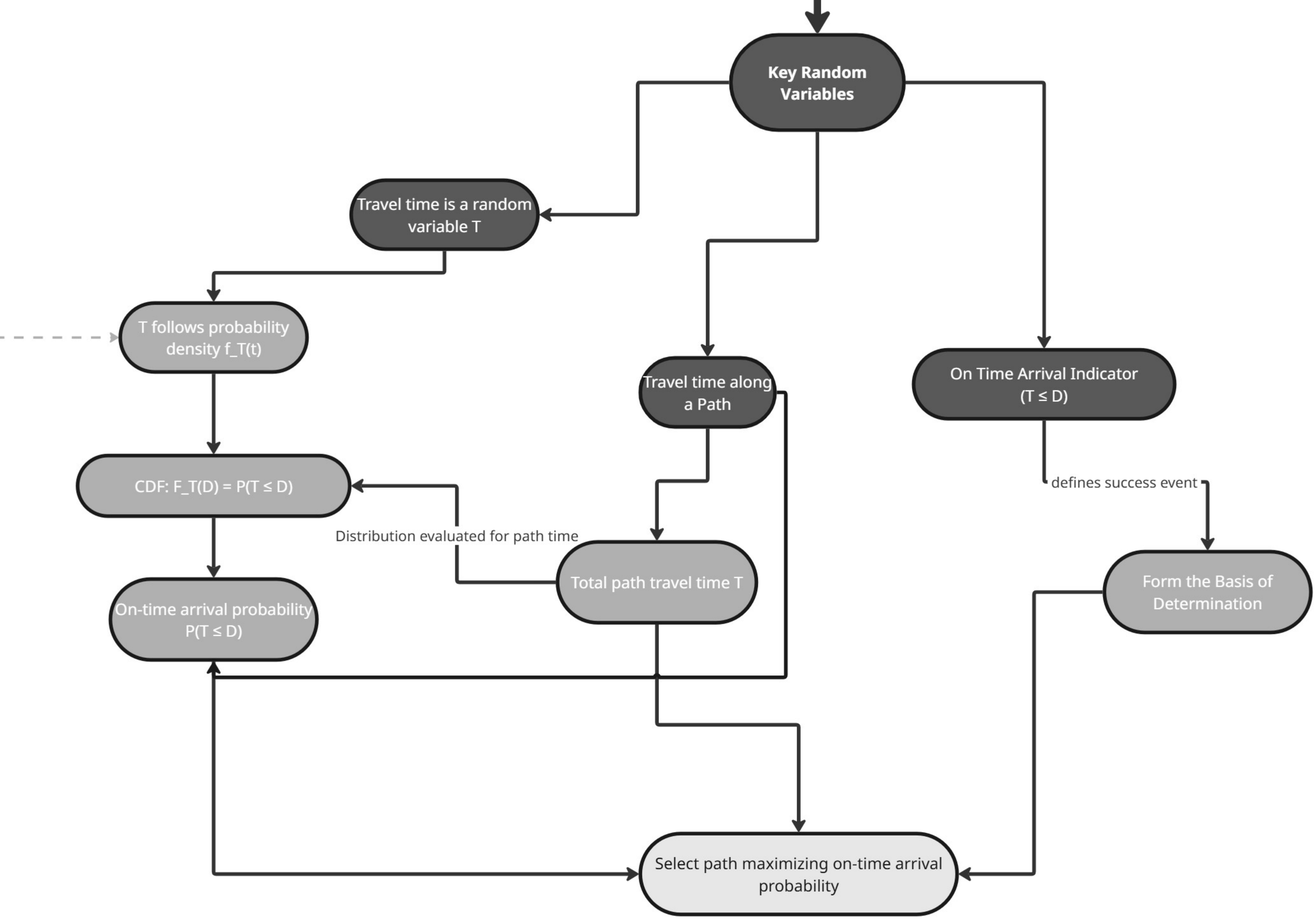
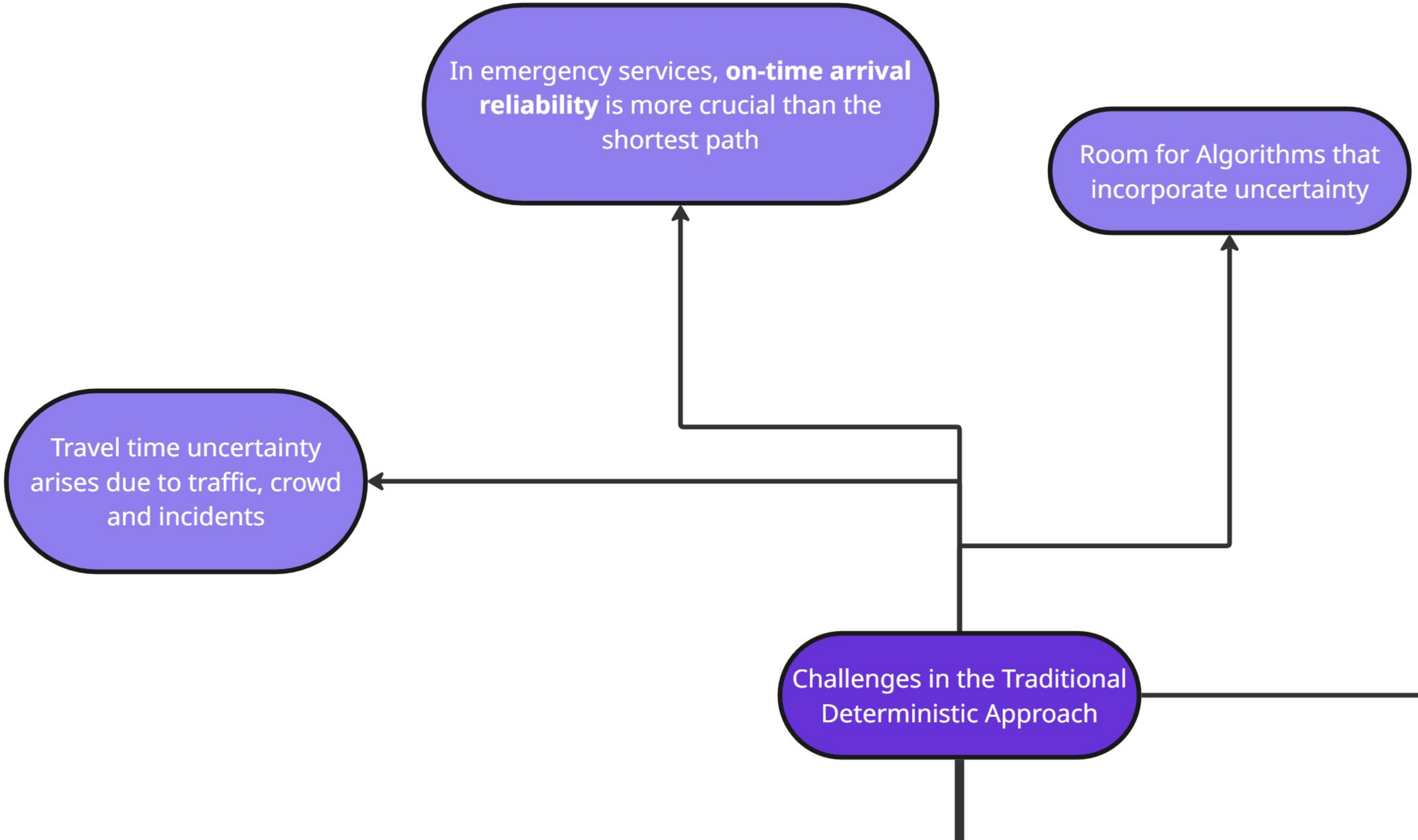
Ignores the Traffic time and the volume of vehicles


Jensen's Inequality Insight
Because of nonlinearity, minimizing E[T] does NOT maximize P(T ≤ D)
KEY PROBLEM:
Deterministic shortest-time routing does not guarantee on-time reliability


Key Random Variables
Travel time is a random variable T
T follows probability density f_T(t)
Travel time along a Path
On Time Arrival Indicator (T ≤ D)
CDF: F_T(D) = P(T ≤ D)
defines success event
Distribution evaluated for path time
Total path travel time T
Form the Basis of Determination
On-time arrival probability P(T ≤ D)
Select path maximizing on-time arrival probability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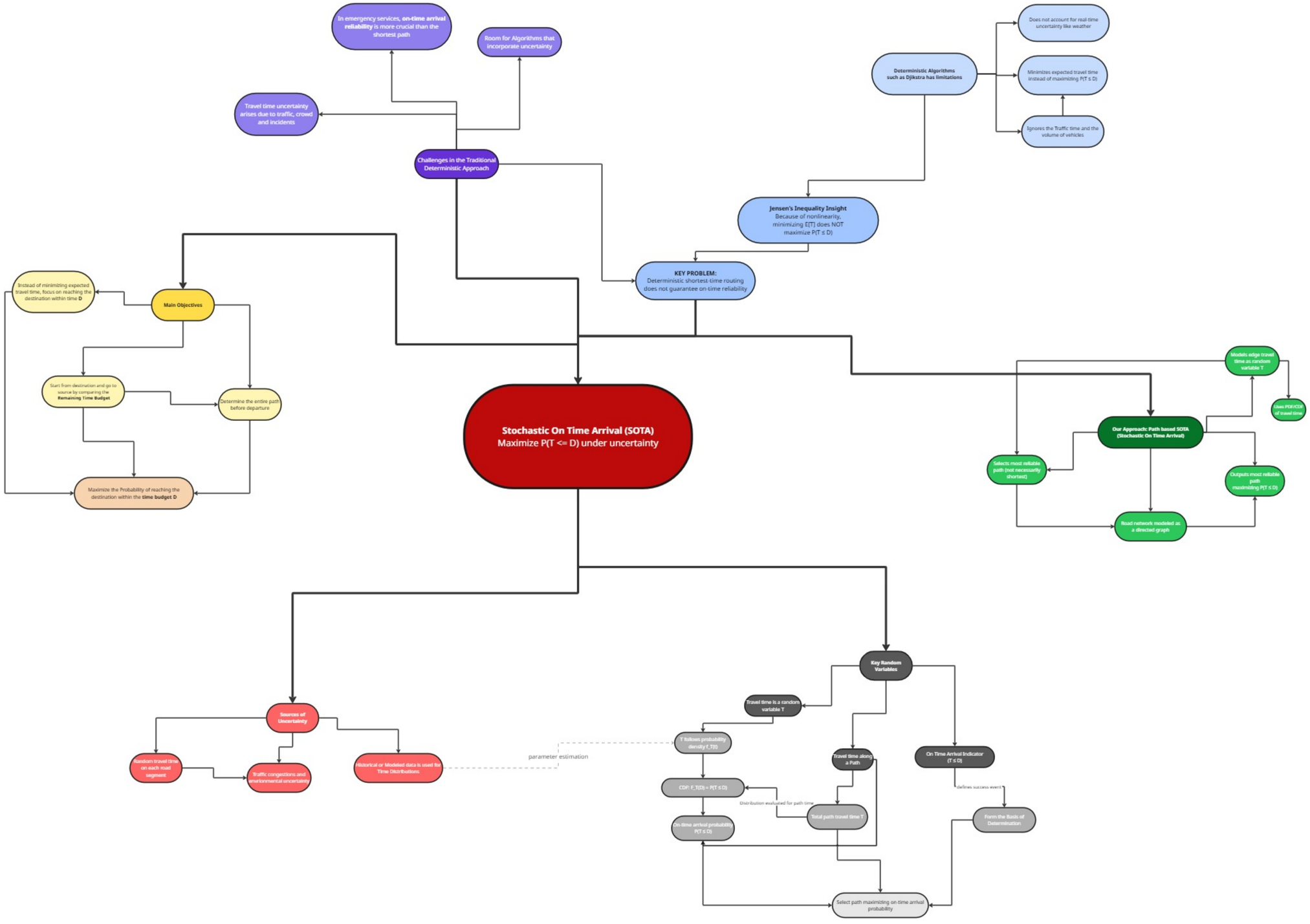
Room for Algorithms that incorporate uncertainty
Travel time uncertainty arises due to traffic, crowd and incidents
Challenges in the Traditional Deterministic Approach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